ENGINEERING PORTFOLIO

Jayden Chung

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I'm a first-year mechanical engineering student at the University of Waterloo and a chassis team member on UW Formula Electric. I learn best hands-on, and I enjoy taking an idea from CAD to a real part I can hold. Working as an assistant automechanic got me interested in cars, but I'm open to any field where I can build things that make people's daily lives better.

Outside of school, I swim, bike and run, and I'm working toward an Ironman one day. Before university, I swam competitively for eight years, racing at the SwimBC Provincial Championships, and gave back to my club by coaching younger swimmers.

Projects

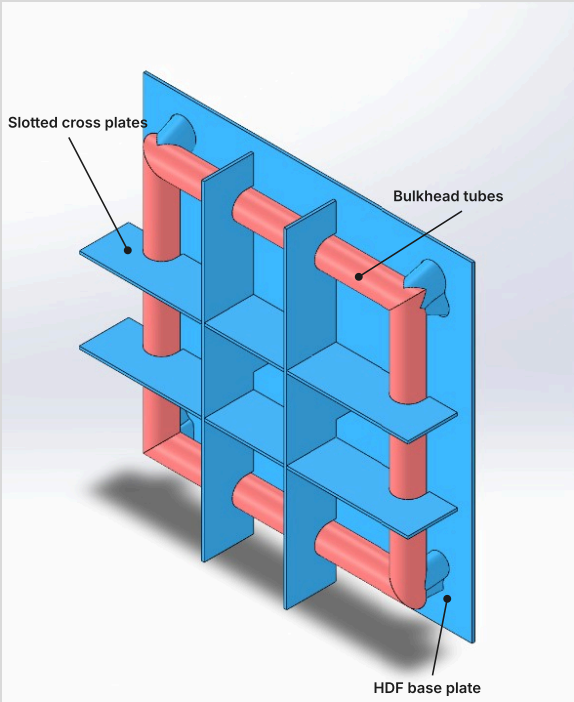
1. Front Bulkhead Weld Jig	UW Formula Electric
2. Kevlar–Balsa Sandwich Dash Panel	UW Formula Electric
3. Carbon Wheelset Upgrade	Personal Project
4. Configurable Brake Balance Bar Trunnion	Coursework

UW FORMULA ELECTRIC · CHASSIS TEAM · OCT. 2026

Front Bulkhead Weld Jig

Designed and built a laser-cut HDF jig that holds the front bulkhead tubes for welding.

Skills: SolidWorks Part and Assembly Modelling · DXF Prep for Laser Cutting · Vertical Bandsaw · Tube Fit-Up



SolidWorks assembly: the jig (blue) with the bulkhead tubes (red).

Why it was needed

The bulkhead tubes have to stay in exact position while they're welded. Without a fixture, they shift and the bulkhead ends up out of shape.

My role

- Designed the slotted cross plates that form the "#" grid, with clearance for laser cutting tolerance and tube fit-up.
- Prepared the DXF files and plate quantities for the Rapid Prototyping Centre.
- Cut the HDF sheet for the base plate down on a vertical bandsaw so it would fit in the laser cutter.
- Assembled the jig and test-fit the frame tubes.

Challenge

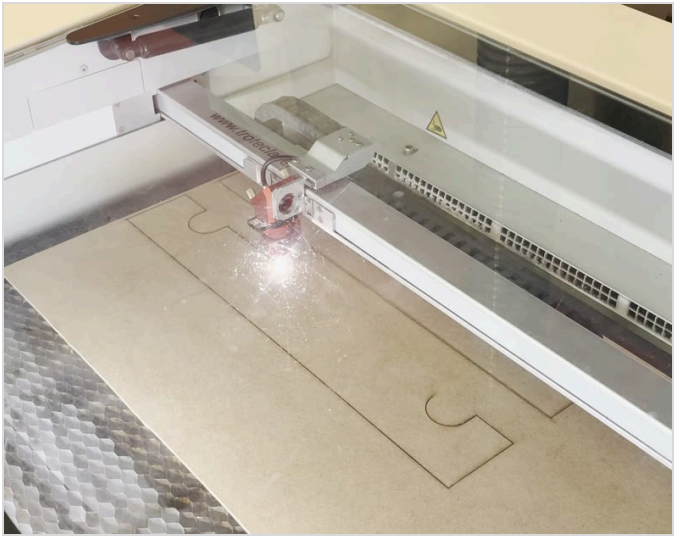
The team needed the jig within two days, and I had only started learning SolidWorks the week before. I picked up reference planes and cut features through my own research and help from the team lead, and built the jig in one day, ahead of the deadline.

Result

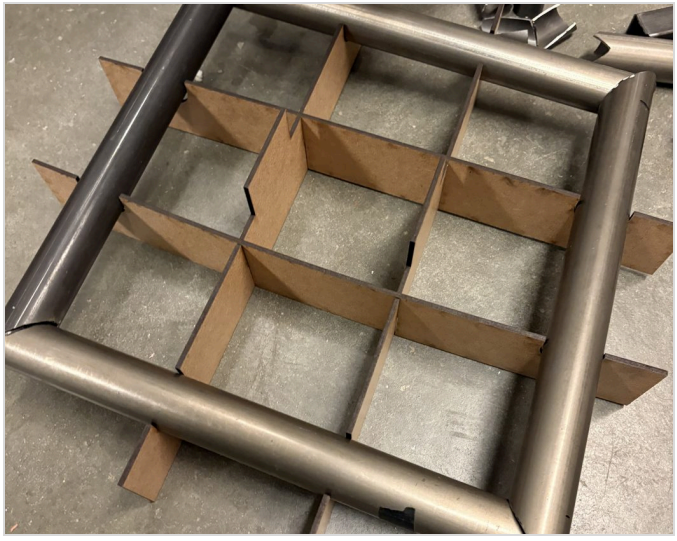
The plates slotted together as modelled. The hand-cut tubes were snug, so I sanded them until they seated.

Next time

I'd measure the hand-cut tubes before finalizing the slot clearances, so they drop in without sanding.



Laser cutting the HDF cross plates.



Assembled grid with the bulkhead tubes test-fit.

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UW FORMULA ELECTRIC · CHASSIS TEAM · SEPT. 2026

Kevlar–Balsa Sandwich Dash Panel

Helped build a Kevlar–balsa test panel to stiffen the car's flexing dashboard.

Skills: Composite Layup · Vacuum Bagging · Epoxy Mixing · Drilling and Fastener Testing · Shop PPE

Why it was needed

Drivers reported that last year's all-Kevlar dashboard flexed when they pressed the buttons. The team wanted to test whether a balsa core between Kevlar skins would stiffen the panel.

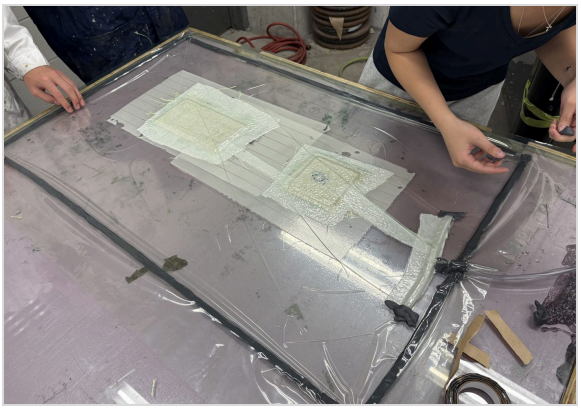
My role

This was my first task on the team, and I helped with every step of the layup:

- Prepped the table by scraping it clean and applying 4 coats of Frekote mould release, 15 minutes apart.
- Cut the Kevlar sheets (3 per side) with all plies at 0° fibre orientation, and weighed them to set the epoxy and hardener quantities.
- Laid up the sheets around the balsa core, then vacuum bagged the panel with peel ply, breather and tacky tape.
- Drilled the cured panel and tightened fasteners through it to check the core for crushing.

Challenge

There were a lot of steps to get right in order. The bag sealed so tightly that the vacuum tubing cut through the plastic, so we patched the leak with putty to hold vacuum.



Panels under vacuum, with peel ply and breather.

Result

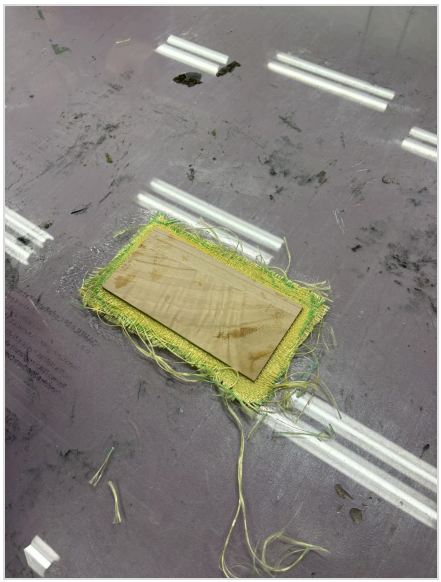
With fasteners tightened hard through it, the core compressed less than last year's all-Kevlar panel. The project and team leads see it as a strong option for the car, pending driver feedback.

Next time

Mix enough epoxy up front. We ran low near the end, and one layer came off after curing.



Cutting Kevlar sheets to size.



Balsa core placed on the Kevlar layers.



Drilling the cured panel for fastener testing.

PERSONAL PROJECT · JULY–AUG. 2026

Carbon Wheelset Upgrade

Sourced and installed a 50 mm deep tubeless carbon wheelset on my Giant TCR, saving about 400 g.



Before: stock wheelset.



After: Elitewheels Ent 2.0 (50 mm) with Pirelli P Zero tires.

What I did

- Compared carbon wheelsets on price, weight and specs such as thru-axle size, and chose the Elitewheels Ent 2.0.
- Sourced the full tubeless setup: tires, rim tape, Stan's tubeless sealant, Presta valves, a sealant injector and tire levers.
- Transferred the cassette, rotors and thru-axles, then checked frame, rotor lockring and cassette clearances.
- Maintain and fit the bike myself: cleaning and lubricating the drivetrain, and angling the handlebars for a more natural hand position.

Challenge

Finding compatible parts took a lot of research, and seating the tight tubeless tires was the hardest step.

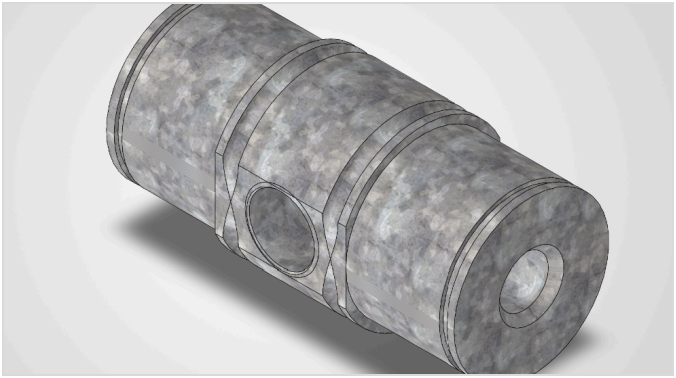
Next time

I missed that my 6-bolt rotors don't fit Centerlock hubs and had to order adapters. Next time I'd check every interface and order all parts at once.

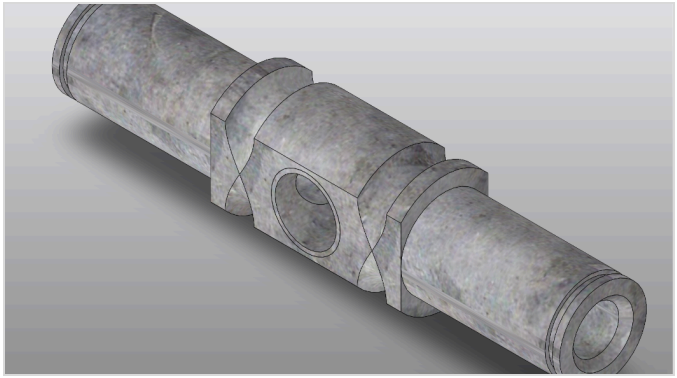
COURSEWORK · OCT. 2026

Configurable Brake Balance Bar Trunnion

Built in SolidWorks by sketching half the profile, revolving it, adding and mirroring the grooves, then cutting the holes with Hole Wizard and extruded cuts. Global variables drive the shaft length and the shaft and brace diameters, so one model covers several configurations.



Short configuration.



Long configuration from the same model.